New L J Institute of Engineering & Technology

SEM: 1st – CSE/CSE (AIML)/DS

Sub: Mathematics-I

Sub Code: BE01R00041

Module 2: Single variable Calculus (Differentiation)

Weightage: 20%

Topics:

- **Chap:1 (5%)**

  Indeterminate Forms and L'Hôspital's Rule.

- **Chap: 2 (15%)**

  Taylor's and Maclaurin's theorem for a function of one variable,

  Taylor's and Maclaurin's series of a function using statement of the theorems,

  Extreme values of functions

❖ Chap 1: Indeterminate Form

We have studied certain rules to evaluate the limit of sum, difference, product & quotient of two functions sometimes become helpless in evaluating the limits of the following type. These limits are known as indeterminate forms. There are seven types of indeterminate forms:

(1) $\lim_{x\to a} \frac{f(x)}{g(x)}$, $if\ \lim_{x\to a} f(x) = 0\ \&\ \lim_{x\to a} g(x) = 0$ $\left(\frac{0}{0}\ form\right)$

(2) $\lim_{x\to a} \frac{f(x)}{g(x)}$, $if\ \lim_{x\to a} f(x) \to \infty\ \&\ \lim_{x\to a} g(x) \to \infty$ $\left(\frac{\infty}{\infty}\ form\right)$

(3) $\lim_{x\to a} f(x).g(x)$, $if\ \lim_{x\to a} f(x) = 0\ \&\ \lim_{x\to a} g(x) \to \infty$ $(0 \times \infty\ form)$

(4) $\lim_{x\to a} f(x) - g(x)$, $if\ \lim_{x\to a} f(x) \to \infty\ \&\ \lim_{x\to a} g(x) \to \infty$ $(\infty - \infty\ form)$

(5) $\lim_{x\to a} f(x)^{g(x)}$, $if\ \lim_{x\to a} f(x) = 0\ \&\ \lim_{x\to a} g(x) = 0$ $(0^0\ form)$

(6) $\lim_{x\to a} f(x)^{g(x)}$, $if\ \lim_{x\to a} f(x) = 1\ \&\ \lim_{x\to a} g(x) \to \infty$ $(1^\infty\ form)$

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(7) $\lim_{x\to a} f(x)^{g(x)}$, $if\ \lim_{x\to a} f(x) \to \infty\ \&\ \lim_{x\to a} g(x) = 0$ $(\infty^0\ form)$

- However, these limits can be solved by using L'Hospital's Rule.

❖ L'Hopistal's Rule:

If $f(x)$ and $g(x)$ are two functions of x which can be expanded by Taylor's series in the neighbourhood of $x = a$ and

if $\lim_{x\to a} f(x) = f(a) = 0,\ \lim_{x\to a} g(x) = g(a) = 0$, then

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$$

Proof:

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f(a) + (x-a)f'(a) + \frac{(x-a)^2 f''(a)}{2!} + \cdots}{g(a) + (x-a)g'(a) + \frac{(x-a)^2 g''(a)}{2!} + \cdots}$$

$$= \lim_{x\to a} \frac{0 + (x-a)f'(a) + \frac{(x-a)^2 f''(a)}{2!} + \cdots}{0 + (x-a)g'(a) + \frac{(x-a)^2 g''(a)}{2!} + \cdots} \qquad \{\because f(a) = g(a) = 0\}$$

$$= \lim_{x\to a} \frac{f'(a) + (x-a)\frac{f''(a)}{2!} + \cdots}{g'(a) + (x-a)\frac{g''(a)}{2!} + \cdots}$$

$$= \frac{f'(a)}{g'(a)}$$

$$= \lim_{x\to a} \frac{f'(x)}{g'(x)} \qquad \{\because If\, g'(a) \neq 0\}$$

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Working Rule:

To Evaluate $\lim_{x\to a} \frac{f(x)}{g(x)}$, $\left(\frac{0}{0} form\right)$

Step 1: Differentiate Numerator & denominator separately & then put $x = a$

Step 2: If again reduces to indeterminate form $\left(\frac{0}{0}\right)$ then again differentiate Numerator & denominator separately & then put $x = a$

Step 3: Process continue till we get finite value of the limit.

Remember Standard Limits:

(1) $\lim_{x\to 0} \frac{\sin x}{x} = 1$ (2) $\lim_{x\to 0} \frac{\tan x}{x} = 1$ (3) $\lim_{x\to 0} \frac{a^x - 1}{x} = \log a$

(4) $\lim_{x\to 0} \frac{e^x - 1}{x} = 1$ (5) $\lim_{x\to 0} (1+x)^{\frac{1}{x}} = e$ (6) $\lim_{x\to \infty} \left(1+\frac{1}{x}\right)^x = e$

(7) $\lim_{x\to 0} \frac{\sin^{-1} x}{x} = 1$ (8) $\lim_{x\to 0} \frac{\sinh x}{x} = 1$ (9) $\lim_{x\to 0} \frac{\tan^{-1} x}{x} = 1$

➢ Type 1: $\frac{0}{0}$ Form

Problems under this type are solved by using L'Hospital's rule considering the fact that

$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$, if $\lim_{x\to a} f(x) = 0$ and $\lim_{x\to a} g(x) = 0$

Example 1: Evaluate $\lim_{x\to 0} \frac{x - \sin x}{x^3}$

Solution:

➢ $\lim_{x\to 0} \frac{x - \sin x}{x^3}$, $\left(\frac{0}{0} form\right)$

$= \lim_{x\to 0} \frac{1 - \cos x}{3x^2}$, $\left(\frac{0}{0}\right)$ [Applying L'Hospital's rule]

$= \lim_{x\to 0} \frac{sinx}{6x}$, $\left(\frac{0}{0}\right)$ [Applying L'Hospital's rule]

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$$= \lim_{x\to 0} \frac{cosx}{6}$$

$$= \frac{1}{6}$$

Example 2: Evaluate $\lim_{x\to 0} \frac{3x-sinx}{x}$

Solution:

- $\lim_{x\to 0} \frac{3x-sinx}{x}$, $\left(\frac{0}{0} form\right)$

$$= \lim_{x\to 0} \frac{3-cosx}{1}$$

$$= \frac{3-1}{1}$$

$$= 2$$

Example 3: Evaluate $\lim_{x\to 0} \frac{e^x-1-x}{x^2}$

Solution:

- $\lim_{x\to 0} \frac{e^x-1-x}{x^2}$, $\left(\frac{0}{0} form\right)$

$$= \lim_{x\to 0} \frac{e^x-1}{2x} \quad , \left(\frac{0}{0}\right)$$

$$= \lim_{x\to 0} \frac{e^x}{2}$$

$$= \frac{1}{2}$$

Example 4: Evaluate $\lim_{x\to 0} \frac{e^x+e^{-x}-x^2-2}{\sin^2 x-x^2}$

Solution:

- $\lim_{x\to 0} \frac{e^x+e^{-x}-x^2-2}{\sin^2 x-x^2}$, $\left(\frac{0}{0} form\right)$

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$$= \lim_{x\to 0} \frac{e^x - e^{-x} - 2x}{2\sin x.\cos x - 2x}$$

$$= \lim_{x\to 0} \frac{e^x - e^{-x} - 2x}{\sin 2x - 2x} \qquad ,\left(\frac{0}{0}\right)$$

$$= \lim_{x\to 0} \frac{e^x + e^{-x} - 2}{2\cos 2x - 2} \qquad ,\left(\frac{0}{0}\right)$$

$$= \lim_{x\to 0} \frac{e^x - e^{-x}}{-4\sin 2x} \qquad ,\left(\frac{0}{0}\right)$$

$$= \lim_{x\to 0} \frac{e^x + e^{-x}}{-8\cos 2x}$$

$$= \frac{1+1}{-8} = -\frac{1}{4}$$

Example 5: Evaluate $\lim_{x\to 0} \frac{tanx - x}{x^2.tanx}$

Solution:

- $\lim_{x\to 0} \frac{tanx - x}{x^2.tanx} \qquad ,\left(\frac{0}{0} form\right)$

$$= \lim_{x\to 0} \frac{tanx - x}{x^2.tanx}\left(\frac{x}{x}\right)$$

$$= \lim_{x\to 0} \frac{tanx - x}{x^3}\left(\frac{x}{tanx}\right)$$

$$= \lim_{x\to 0} \frac{tanx - x}{x^3}\left(\frac{1}{\frac{tanx}{x}}\right) \qquad ,\left[\because \lim_{x\to 0}\frac{tanx}{x} = 1\right]$$

$$= \lim_{x\to 0} \frac{tanx - x}{x^3}(1) \qquad ,\left(\frac{0}{0}\right)$$

$$= \lim_{x\to 0} \frac{sec^2x - 1}{3x^2} \qquad ,\left(\frac{0}{0}\right)$$

$$= \lim_{x\to 0} \frac{2secx(secx.tanx)}{6x} \qquad ,\left(\frac{0}{0}\right)$$

$$= \lim_{x\to 0} \frac{2sec^2x}{6}.\left(\frac{tanx}{x}\right)$$

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$$= \lim_{x\to 0} \frac{sec^2 x}{3}.(1)$$

$$= \frac{1}{3}$$

Example 6: Evaluate $\lim_{x\to 0} \frac{tan^2 x - x^2}{x^2 tan^2 x}$

Solution:

- $\lim_{x\to 0} \frac{tan^2 x - x^2}{x^2 tan^2 x}$, $\left(\frac{0}{0} form\right)$

$$= \lim_{x\to 0} \frac{tan^2 x - x^2}{x^2 tan^2 x}\left(\frac{x^2}{x^2}\right)$$

$$= \lim_{x\to 0} \frac{tan^2 x - x^2}{x^4}\left(\frac{x^2}{tan^2 x}\right)$$

$$= \lim_{x\to 0} \frac{tan^2 x - x^2}{x^4}\left(\frac{1}{\frac{tan^2 x}{x^2}}\right) \quad , \left[\because \lim_{x\to 0} \frac{tan^2 x}{x^2} = 1\right]$$

$$= \lim_{x\to 0} \frac{tan^2 x - x^2}{x^4}(1) \quad , \left(\frac{0}{0}\right)$$

$$= \lim_{x\to 0} \frac{2tanx.sec^2 x - 2x}{4x^3} \quad , \left(\frac{0}{0}\right)$$

$$= \lim_{x\to 0} \frac{2tanx(1 + tan^2 x) - 2x}{4x^3} \quad , \left(\frac{0}{0}\right)$$

$$= \lim_{x\to 0} \frac{1}{2}\left[\frac{tanx + tan^3 x - x}{x^3}\right] \quad , \left(\frac{0}{0}\right)$$

$$= \lim_{x\to 0} \frac{1}{2}\left[\frac{tanx - x}{x^3}\right] + \lim_{x\to 0} \frac{1}{2}\left[\frac{tan^3 x}{x^3}\right]$$

$$= \lim_{x\to 0} \frac{1}{2}\left[\frac{sec^2 x - 1}{3x^2}\right] + \lim_{x\to 0} \frac{1}{2}.(1)$$

$$= \lim_{x\to 0} \frac{1}{2}\left[\frac{2secx(secx.tanx) - 1}{6x}\right] + \frac{1}{2}$$

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$$= \lim_{x\to 0} \frac{1}{\cancel{2}}\left[\frac{\cancel{2}sec^2x}{6}.\left(\frac{tanx}{x}\right)\right] + \frac{1}{2}$$

$$= \left(\frac{1}{6}+\frac{1}{2}\right) = \frac{1+3}{6} = \frac{4}{6}$$

$$= \frac{2}{3}$$

Example 7: Evaluate $\lim_{x\to 1} \frac{x^x - x}{x-1-logx}$

Solution:

- $\lim_{x\to 1} \frac{x^x - x}{x-1-logx}$, $\left(\frac{0}{0} form\right)$

$$= \lim_{x\to 1} \frac{x^x(1+logx)-1}{1-\frac{1}{x}} \quad , \left(\frac{0}{0}\right)$$

$$= \lim_{x\to 1} \frac{x^x\left(\frac{1}{x}\right) + x^x(1+logx)^2 - 1}{\frac{1}{x^2}}$$

$$= \frac{1+1}{1}$$

$$= 2$$

Example 8: Evaluate $\lim_{x\to \frac{1}{2}} \frac{cos^2\pi x}{e^{2x}-2ex}$

Solution:

- $\lim_{x\to \frac{1}{2}} \frac{cos^2\pi x}{e^{2x}-2ex}$, $\left(\frac{0}{0} form\right)$

$$= \lim_{x\to \frac{1}{2}} \frac{2cos\pi x(-sin\pi x)(\pi)}{2e^{2x}-2e}$$

$$= \lim_{x\to \frac{1}{2}} \frac{cos\pi x(-sin\pi x)(\pi)}{e^{2x}-e} \quad , \left(\frac{0}{0}\right)$$

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$$= \lim_{x\to\frac{1}{2}} \pi.\left[\frac{cos\pi x(-cos\pi x)(\pi) + (-sin\pi x)(-sin\pi x)\pi}{2e^{2x}}\right]$$

$$= \pi^2.\lim_{x\to\frac{1}{2}} \frac{-cos^2\pi x + sin^2\pi x}{2e^{2x}}$$

$$= \frac{\pi^2(0+1)}{2e}$$

$$= \frac{\pi^2}{2e}$$

Example 9: Evaluate $\lim_{x\to\frac{\pi}{2}} \frac{log(sinx)}{(\pi-2x)^2}$

Solution:

- $\lim_{x\to\frac{\pi}{2}} \frac{log(sinx)}{(\pi-2x)^2}$, $\left(\frac{0}{0} form\right)$

$$= \lim_{x\to\frac{\pi}{2}} \frac{\frac{cosx}{sinx}}{2(\pi-2x)(-2)}$$

$$= \lim_{x\to\frac{\pi}{2}} \frac{cotx}{-4(\pi-2x)} \quad , \left(\frac{0}{0}\right)$$

$$= \lim_{x\to\frac{\pi}{2}} \frac{-cosec^2x}{-4(-2)}$$

$$= -\frac{1}{8}$$

Example 10: Evaluate $\lim_{x\to 0} \frac{xe^x - log(1+x)}{x^2}$

Solution:

- $\lim_{x\to 0} \frac{xe^x - log(1+x)}{x^2}$, $\left(\frac{0}{0} form\right)$

$$= \lim_{x\to 0} \frac{e^x + xe^x - \frac{1}{1+x}}{2x} \quad , \left(\frac{0}{0}\right)$$

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$$= \lim_{x\to 0} \frac{e^x + e^x + xe^x + \frac{1}{(1+x)^2}}{2}$$

$$= \frac{3}{2}$$

Example 11: Evaluate $\lim_{x\to 0} \frac{e^x - e^{-x} - 2\log(1+x)}{x\, sinx}$

Solution:

- $\lim_{x\to 0} \frac{e^x - e^{-x} - 2\log(1+x)}{x\, sinx}$, $\left(\frac{0}{0} form\right)$

$$= \lim_{x\to 0} \frac{e^x + e^{-x} - 2\left(\frac{1}{1+x}\right)}{sinx + x.cosx} \quad , \left(\frac{0}{0}\right)$$

$$= \lim_{x\to 0} \frac{e^x - e^{-x} + \frac{2}{(1+x)^2}}{cosx + cosx - x\, sinx}$$

$$= \frac{2}{2}$$

$$=1$$

Example 12: Evaluate $\lim_{x\to 0} \frac{e^x - e^{-x} - 2x}{x - sinx}$

Solution:

- $\lim_{x\to 0} \frac{e^x - e^{-x} - 2x}{x - sinx}$, $\left(\frac{0}{0} form\right)$

$$= \lim_{x\to 0} \frac{e^x + e^{-x} - 2}{1 - cosx} \quad , \left(\frac{0}{0}\right)$$

$$= \lim_{x\to 0} \frac{e^x - e^{-x}}{sinx} \quad , \left(\frac{0}{0}\right)$$

$$= \lim_{x\to 0} \frac{e^x + e^{-x}}{cosx}$$

$$= \frac{1+1}{1}$$

$$= 2$$

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H.W.

Example 13: Evaluate $\lim_{x\to 0}\frac{x-tanx}{x^3}$ **ANS:** $-\frac{1}{3}$

Example 14: Evaluate $\lim_{x\to 0}\frac{x^2+2cosx-2}{x.sin^3x}$ **ANS:** $\frac{1}{12}$

Example 15: Evaluate $\lim_{x\to 0}\frac{2x-xcosx-sinx}{2x^3}$ **ANS:** $\frac{1}{3}$

Example 16: Find $\lim_{x\to a}\frac{\sin x-\sin a}{(x-a)^2}$ **ANS: 0**

➢ Type 2: $\frac{\infty}{\infty}$ Forms

If $\lim_{x\to a} f(x)\to\infty$ & $\lim_{x\to a} g(x)\to\infty$ then $\lim_{x\to a}\frac{f(x)}{g(x)}$ becomes $\frac{\infty}{\infty}$ Form.

$$\lim_{x\to a}\frac{f(x)}{g(x)}=\lim_{x\to a}\frac{\frac{1}{g(x)}}{\frac{1}{f(x)}}\ \left(\frac{0}{0}form\right)$$

Example 1: Evaluate $\lim_{x\to\frac{\pi}{2}}\frac{secx}{1+tanx}$

Solution:

➢ $\lim_{x\to\frac{\pi}{2}}\frac{secx}{1+tanx}$, $\left(\frac{\infty}{\infty}form\right)$

$$=\lim_{x\to\frac{\pi}{2}}\frac{secx.tanx}{3sec^2x}$$

$$=\lim_{x\to\frac{\pi}{2}}\frac{tanx}{secx}$$

$$=\lim_{x\to\frac{\pi}{2}}\frac{sinx}{cosx}.cosx$$

$$=\lim_{x\to\frac{\pi}{2}}sinx\ =1$$

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Example 2: Evaluate $\lim_{x\to 0} \frac{logsinx}{cotx}$

Solution:

- $\lim_{x\to 0} \frac{logsinx}{cotx}$ $,\left(\frac{\infty}{\infty} form\right)$

$$= \lim_{x\to 0} \frac{\frac{1}{sinx}.cosx}{-cosec^2x}$$

$$= \lim_{x\to 0} -sinx.cosx$$

$$= 0$$

Example 3: Evaluate $\lim_{x\to a} \frac{log(x-a)}{log(e^x-e^a)}$

Solution:

- $\lim_{x\to a} \frac{log(x-a)}{log(e^x-e^a)}$ $,\left(\frac{\infty}{\infty} form\right)$

$$= \lim_{x\to a} \frac{\frac{1}{(x-a)}.1}{\frac{1}{e^x-e^a}.e^x}$$

$$= \lim_{x\to a} \frac{e^x-e^a}{(x-a)e^x} \quad ,\left(\frac{0}{0}\right)$$

$$= \lim_{x\to a} \frac{e^x}{(x-a)e^x+1.e^x} \quad ,\left(\frac{0}{0}\right)$$

$$= \frac{e^a}{0+e^a}$$

$$= 1$$

Example 4: Evaluate $\lim_{x\to 0} \frac{cotx}{cot2x}$

Solution:

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- $\lim_{x\to 0} \frac{cotx}{cot2x}$, $\left(\frac{\infty}{\infty} form\right)$

$= \lim_{x\to 0} \frac{tan2x}{tanx}$, $\left(\frac{0}{0} form\right)$

$= \lim_{x\to 0} \frac{2sec^2 2x}{sec^2 x}$

$= \lim_{x\to 0} \frac{2cos^2 x}{cos^2 2x}$

$= 2$

Example 5: Evaluate $\lim_{x\to 0} \log_{tanx} tan2x$

Solution: Changing the base to e

- $\lim_{x\to 0} \frac{\log tan2x}{logtanx}$, $\left(\frac{\infty}{\infty} form\right) \left[\because \log_a b = \frac{\log_e b}{\log_e a}\right]$

$= \lim_{x\to 0} \frac{\frac{1}{tan2x}(2sec^2 2x)}{\frac{1}{tanx}(sec^2 x)}$

$= \lim_{x\to 0} 2\left(\frac{tanx}{tan2x}\right)\left(\frac{sec^2 2x}{sec^2 x}\right)$

$= \lim_{x\to 0} \left(\frac{tanx}{x}\right)\left(\frac{2x}{tan2x}\right)\left(\frac{sec^2 2x}{sec^2 x}\right)$

$= 1.1.1$, $\left[\because \lim_{x\to 0} \frac{tanx}{x} = 1\right]$

$= 1$

H.W.

Example 6: Evaluate $\lim_{x\to \frac{\pi}{2}} \frac{log\left(x-\frac{\pi}{2}\right)}{tanx}$ **ANS: 0**

Example 7: Evaluate $\lim_{x\to 0} \frac{ln(sinx)}{\left(\frac{1}{x}\right)}$ **ANS: 0**

Example 8: Evaluate $\lim_{x\to \infty} \frac{x-2x^2}{3x^2+5x}$ **ANS: $-\frac{2}{3}$**

Example 9: Evaluate $\lim_{x\to 0} \log_x tanx$ **ANS: 1**

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➢ Type 3: $0 \times \infty$ Form

If $\lim_{x\to a} f(x) = 0$ & $\lim_{x\to a} g(x) \to \infty$ then $\lim_{x\to a} f(x).g(x)$ becomes $0 \times \infty$ Form.

$$\lim_{x\to a} f(x).g(x) = \lim_{x\to a} \frac{f(x)}{\frac{1}{g(x)}} \left(\frac{0}{0} \; or \; \frac{\infty}{\infty} \; form\right)$$

Example 1: Evaluate $\lim_{x\to\frac{1}{2}} \left(x^2 - \frac{1}{4}\right).tan\pi x$

Solution:

➢ $\lim_{x\to\frac{1}{2}} \left(x^2 - \frac{1}{4}\right).tan\pi x$, $(0 \times \infty \; form)$

$$= \lim_{x\to\frac{1}{2}} \frac{x^2 - \frac{1}{4}}{cot\pi x} \quad , \left(\frac{0}{0} form\right)$$

$$= \lim_{x\to\frac{1}{2}} \frac{2x}{\pi(-cosec^2\pi x)}$$

$$= \frac{2\left(\frac{1}{2}\right)}{-\pi(1)^2}$$

$$= -\frac{1}{\pi}$$

Example 2: Evaluate $\lim_{x\to\frac{\pi}{4}} (1 - tanx).sec2x$

Solution:

➢ $\lim_{x\to\frac{\pi}{4}} (1 - tanx).sec2x$, $(0 \times \infty \; form)$

$$= \lim_{x\to\frac{\pi}{4}} \frac{1 - tanx}{cos2x} \quad , \left(\frac{0}{0} form\right)$$

$$= \lim_{x\to\frac{\pi}{4}} \frac{-sec^2x}{-2.\sin 2x}$$

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$$= \frac{(\sqrt{2})^2}{2}$$

$$= 1$$

Example 3: Evaluate $\lim_{x\to a} log\left(2-\frac{x}{a}\right).cot(x-a)$

Solution:

- $\lim_{x\to a} log\left(2-\frac{x}{a}\right).cot(x-a)$, $(0\times\infty form)$

$$= \lim_{x\to a}\frac{log\left(2-\frac{x}{a}\right)}{tan(x-a)} \quad , \left(\frac{0}{0} form\right)$$

$$= \lim_{x\to a}\frac{log(2a-x)-loga}{tan(x-a)} \quad , \left(\frac{0}{0}\right)$$

$$= \lim_{x\to a}\frac{\frac{1}{2a-x}(-1)}{sec^2(x-a)(1)}$$

$$= -\frac{1}{a}$$

Example 4: Evaluate $\lim_{v\to\infty} v^2\left[1-e^{-\frac{2gx}{v^2}}\right]$

Solution:

- $\lim_{v\to\infty} v^2\left[1-e^{-\frac{2gx}{v^2}}\right]$, $(0\times\infty form)$

Let $\frac{1}{v^2}=t$, When $v\to\infty \implies t\longrightarrow 0$

$$= \lim_{t\to 0}\frac{[1-e^{-2gxt}]}{t} \quad , \left(\frac{0}{0} form\right)$$

$$= \lim_{t\to 0}\frac{[0-e^{-2gxt}(-2gx)]}{1}$$

$$= 2gx$$

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Example 5: Evaluate $\lim_{x\to 0} x.\log x$

Solution:

- $\lim_{x\to 0} x.\log x$, $(0 \times \infty\ form)$

$= \lim_{x\to 0} \frac{logx}{\frac{1}{x}}$, $\left(\frac{\infty}{\infty} form\right)$

$= \lim_{x\to 0} \frac{\frac{1}{x}}{-\frac{1}{x^2}}$

$= \lim_{x\to 0} (-x)$

$= 0$

Example 6: Evaluate $\lim_{x\to\infty} \left(a^{\frac{1}{x}} - 1\right) x$

Solution:

- Here Take $\frac{1}{x} = t$
- When $x \to \infty \Longrightarrow t \longrightarrow 0$
- $\lim_{x\to\infty} \left(a^{\frac{1}{x}} - 1\right) x$, $(0 \times \infty\ form)$

$= \lim_{t\to 0} (a^t - 1) \frac{1}{t}$, $\left(\frac{0}{0}\ form\right)$

$= \lim_{t\to 0} \frac{a^t\, loga}{1} = log\ a$

H.W.

Example 7: Evaluate $\lim_{x\to 1} (x^2 - 1).tan\frac{\pi x}{2}$ **ANS: $-\frac{4}{\pi}$**

Example 8: Evaluate $\lim_{x\to 0} sinx.logx$ **ANS: 0**

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➢ Type 4: $\infty - \infty$ Form

If $\lim_{x\to a} f(x) \to \infty$ & $\lim_{x\to a} g(x) \to \infty$ then $\lim_{x\to a} f(x) - g(x)$ becomes $\infty - \infty$ Form.

$$\lim_{x\to a} f(x) - g(x) = \lim_{x\to a} \frac{\frac{1}{g(x)} - \frac{1}{f(x)}}{\frac{1}{f(x).g(x)}} \left(\frac{0}{0} \text{ or } \frac{\infty}{\infty} form\right)$$

Example 1: Evaluate $\lim_{x\to 0} \frac{1}{sinx} - \frac{1}{x}$

Solution:

➢ $\lim_{x\to 0} \frac{1}{sinx} - \frac{1}{x}$, $(\infty - \infty\ form)$

$= \lim_{x\to 0} \frac{x - sinx}{x.sinx}$, $\left(\frac{0}{0} form\right)$

$= \lim_{x\to 0} \frac{1 - cosx}{sinx + x.cosx}$, $\left(\frac{0}{0}\right)$

$= \lim_{x\to 0} \frac{sinx}{\text{cosx} + (\text{cosx} - \text{xsinx})}$

$= \frac{0}{2}$

$= 0$

Example 2: Evaluate $\lim_{x\to 0} \frac{1}{x^2} - \frac{1}{sin^2 x}$

Solution:

➢ $\lim_{x\to 0} \frac{1}{x^2} - \frac{1}{sin^2 x}$, $(\infty - \infty\ form)$

$= \lim_{x\to 0} \frac{sin^2 x - x^2}{x^2.sin^2 x}$, $\left(\frac{0}{0} form\right)$

$= \lim_{x\to 0} \frac{sin^2 x - x^2}{x^2.sin^2 x}\left(\frac{x^2}{x^2}\right)$

$= \lim_{x\to 0} \frac{sin^2 x - x^2}{x^4}\left(\frac{x^2}{sin^2 x}\right)$

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$$= \lim_{x\to 0} \frac{sin^2x - x^2}{x^4}(1) \qquad ,\left(\frac{0}{0}\right)$$

$$= \lim_{x\to 0} \frac{2sinxcosx - 2x}{4x^3}$$

$$= \lim_{x\to 0} \frac{sin2x - 2x}{4x^3} \qquad ,\left(\frac{0}{0}\right)$$

$$= \lim_{x\to 0} \frac{2cos2x - 2}{12x^2} \qquad ,\left(\frac{0}{0}\right)$$

$$= \lim_{x\to 0} \frac{-4sin2x}{24x} \qquad ,\left(\frac{0}{0}\right)$$

$$= \lim_{x\to 0} \left(\frac{-4}{24}\right)\left(\frac{sin2x}{2x}\right)$$

$$= \lim_{x\to 0} \left(\frac{-1}{3}\right)(1)$$

$$= \frac{-1}{3}$$

Example 3: Evaluate $\lim_{x\to 0} \frac{1}{x} - \frac{1}{tanx}$ or $\lim_{x\to 0} \frac{1}{x} - cotx$ or $\lim_{x\to 0} \frac{1}{x}(1 - x.cotx)$

Solution:

- $\lim_{x\to 0} \frac{1}{x} - \frac{1}{tanx} \qquad ,(\infty - \infty\ form)$

$$= \lim_{x\to 0} \frac{tanx - x}{x.tanx} \qquad ,\left(\frac{0}{0} form\right)$$

$$= \lim_{x\to 0} \frac{tanx - x}{x.tanx}\left(\frac{x}{x}\right)$$

$$= \lim_{x\to 0} \frac{tanx - x}{x^2}\left(\frac{x}{tanx}\right)$$

$$= \lim_{x\to 0} \frac{tanx - x}{x^2}(1) \qquad ,\left(\frac{0}{0}\right)$$

$$= \lim_{x\to 0} \frac{sec^2x - 1}{2x} \qquad ,\left(\frac{0}{0}\right)$$

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$$= \lim_{x\to 0} \frac{2secx(sexc.tanx)}{2} \quad ,\left(\frac{0}{0}\right)$$

$$= \lim_{x\to 0} sec^2x.tanx$$

$$= (1).(0)$$

$$= 0$$

Example 4: Evaluate $\lim_{x\to 1}\left(\frac{x}{x-1} - \frac{1}{logx}\right)$

Solution:

- $\lim_{x\to 1}\left(\frac{x}{x-1} - \frac{1}{logx}\right) \quad ,(\infty - \infty\ form)$

$$= \lim_{x\to 1} \frac{x.logx - x + 1}{(x-1).logx} \quad ,\left(\frac{0}{0} form\right)$$

$$= \lim_{x\to 1} \frac{\left[x\left(\frac{1}{x}\right) + 1.logx\right] - 1}{\left[(x-1)\left(\frac{1}{x}\right) + 1.logx\right]}$$

$$= \lim_{x\to 1} \frac{logx}{1 - \frac{1}{x} + logx} \quad ,\left(\frac{0}{0}\right)$$

$$= \lim_{x\to 1} \frac{\frac{1}{x}}{-\left(-\frac{1}{x^2}\right) + \frac{1}{x}}$$

$$= \frac{1}{1+1}$$

$$= \frac{1}{2}$$

Example 5: Evaluate $\lim_{x\to 0} \frac{1}{x} - \frac{1}{x^2} log(1+x)$

Solution:

- $\lim_{x\to 0} \frac{1}{x} - \frac{1}{x^2} log(1+x) \quad ,(\infty - \infty\ form)$

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$$= \lim_{x\to 0} \frac{x - log(1+x)}{x^2} \qquad , \left(\frac{0}{0} form\right)$$

$$= \lim_{x\to 0} \frac{1 - \frac{1}{1+x}}{2x} \qquad , \left(\frac{0}{0}\right)$$

$$= \lim_{x\to 0} \frac{-\left[-\frac{1}{(1+x)^2}\right]}{2}$$

$$= \frac{1}{2}$$

H.W.

Example 6: Evaluate $\lim_{x\to 0} (cosecx - cotx)$ **ANS: 0**

Example 7: Evaluate $\lim_{x\to 0} \left[\frac{\pi}{4x} - \frac{\pi}{2x(e^{\pi x}+1)}\right]$ **ANS: $\frac{\pi^2}{8}$**

➢ Type 5, 6 & 7: $0^0, 1^\infty$ & ∞^0 Form

If $\lim_{x\to a} f(x)^{g(x)}$ leads to any one of the indeterminate forms $0^0, 1^\infty$ & ∞^0 then following procedure for their evaluation.

$\Longrightarrow Put\ y = f(x)^{g(x)}$

$\Longrightarrow$ Taking log on both the sides,

$\Longrightarrow logy = g(x).logf(x)$

$\Longrightarrow Let\ l = \lim_{x\to a} logy = \lim_{x\to a} g(x).logf(x)$ then,

$\Longrightarrow \lim_{x\to a} y = e^l$

$\Longrightarrow \lim_{x\to a} f(x)^{g(x)} = e^l$

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Example 1: Evaluate $\lim_{x\to 0} sinx^{tanx}$, $(0^0\ form)$

Solution:

- $y = sinx^{tanx}$ **,** Now take log on both the sides,
- $logy = tanx.\log(sinx)$
- $\lim_{x\to 0} logy = \lim_{x\to 0} tanx.\log(sinx)$, $(0 \times \infty\ form)$

$$= \lim_{x\to 0} \frac{log(sinx)}{cotx} \quad , \left(\frac{\infty}{\infty} form\right)$$

$$= \lim_{x\to 0} \frac{\frac{1}{sinx}.cosx}{-cosec^2x}$$

$$= \lim_{x\to 0} -cosx.sinx$$

$$= -(1)(0)$$

$\therefore \lim_{x\to 0} logy = 0$

$\therefore \lim_{x\to 0} y = e^0 = 1$

$\therefore \lim_{x\to 0} sinx^{tanx} = e^0 = 1$

Example 2: Evaluate $\lim_{x\to 1} (1-x^2)^{\frac{1}{log(1-x)}}$, $(0^0\ form)$

Solution:

- $y = (1-x^2)^{\frac{1}{log(1-x)}}$ **,** Now take log on both the sides,
- $logy = \frac{log(1-x^2)}{\log(1-x)}$
- $\lim_{x\to 1} logy = \lim_{x\to 1} \frac{log(1-x^2)}{\log(1-x)}$, $\left(\frac{\infty}{\infty} form\right)$

$$= \lim_{x\to 0} \frac{\frac{-2x}{1-x^2}}{\frac{-1}{1-x}}$$

$$= \lim_{x\to 1} \frac{2x(1-x)}{(1-x^2)}$$

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$$= \lim_{x\to 1} \frac{2x(1-x)}{(1-x)(1+x)}$$

$$= \lim_{x\to 1} \frac{2x}{(1+x)}$$

$$= \frac{2}{1+1}$$

$$\therefore \lim_{x\to 1} logy = 1$$

$$\therefore \lim_{x\to 1} y = e^1 = e$$

$$\therefore \lim_{x\to 1} (1-x^2)^{\frac{1}{log(1-x)}} = e$$

Example 3: Evaluate $\lim_{x\to\frac{\pi}{2}} (cosx)^{\frac{\pi}{2}-x}$, $(0^0\ form)$

Solution:

- $y = (cosx)^{\frac{\pi}{2}-x}$, Now take log on both the sides,
- $logy = \left(\frac{\pi}{2} - x\right) log(cosx)$
- $\lim_{x\to\frac{\pi}{2}} logy = \lim_{x\to\frac{\pi}{2}} \left(\frac{\pi}{2} - x\right) log(cosx)$, $(0 \times \infty form)$

$$= \lim_{x\to\frac{\pi}{2}} \frac{log(cosx)}{\frac{1}{\left(\frac{\pi}{2}-x\right)}} \quad , \left(\frac{\infty}{\infty} form\right)$$

$$= \lim_{x\to\frac{\pi}{2}} \frac{-\frac{sinx}{cosx}}{\frac{-1}{\left(\frac{\pi}{2}-x\right)^2}(-1)}$$

$$= \lim_{x\to\frac{\pi}{2}} -tanx.\left(\frac{\pi}{2} - x\right)^2 \quad , (0 \times \infty form)$$

$$= \lim_{x\to\frac{\pi}{2}} \frac{-\left(\frac{\pi}{2}-x\right)^2}{cotx} \quad , \left(\frac{0}{0} form\right)$$

$$= \lim_{x\to\frac{\pi}{2}} \frac{-2\left(\frac{\pi}{2} - x\right)(-1)}{-cosec^2x}$$

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$$= \lim_{x\to\frac{\pi}{2}} -2\left(\frac{\pi}{2}-x\right)sin^2x$$

$$= -2(0)(1)$$

$$= 0$$

$$\therefore \lim_{x\to\frac{\pi}{2}} logy = 0$$

$$\therefore \lim_{x\to\frac{\pi}{2}} y = e^0 = 1$$

$$\therefore \lim_{x\to\frac{\pi}{2}} (cosx)^{\frac{\pi}{2}-x} = 1$$

Example 4: Evaluate $\lim_{x\to 0}\left(\frac{a^x+b^x+c^x}{3}\right)^{\frac{1}{3x}}$, $(1^\infty\ form)$

Solution:

- $y = \left(\frac{a^x+b^x+c^x}{3}\right)^{\frac{1}{3x}}$ **,** Now take log on both the sides,
- $logy = \frac{1}{3x} log\left(\frac{a^x+b^x+c^x}{3}\right)$
- $\lim_{x\to 1} logy = \lim_{x\to 0} \frac{log\left(\frac{a^x+b^x+c^x}{3}\right)}{3x}$, $\left(\frac{0}{0}form\right)$

$$= \lim_{x\to 0} \frac{log(a^x+b^x+c^x)-log3}{3x} \quad , \left(\frac{0}{0}form\right)$$

$$= \lim_{x\to 0} \frac{\frac{1}{a^x+b^x+c^x}(a^x loga+b^x logb+c^x logc)}{3}$$

$$= \frac{1}{(1+1+1)} \frac{(loga+logb+logc)}{3}$$

$$= \frac{1}{3}\frac{log(abc)}{3}$$

$$\therefore \lim_{x\to 0} logy = \log(abc)^{\frac{1}{9}}$$

$$\therefore \lim_{x\to 0} y = (abc)^{\frac{1}{9}}$$

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$$\therefore \lim_{x\to 0}\left(\frac{a^x+b^x+c^x}{3}\right)^{\frac{1}{3x}} = (abc)^{\frac{1}{9}}$$

Example 5: Evaluate $\lim_{x\to 0}(a^x+x)^{\frac{1}{x}}$, $(1^\infty\ form)$

Solution:

- $y=(a^x+x)^{\frac{1}{x}}$ **,** Now take log on both the sides,
- $logy = \frac{1}{x}log(a^x+x)$
- $\lim_{x\to 1} logy = \lim_{x\to 0}\frac{log(a^x+x)}{x}$, $\left(\frac{0}{0}form\right)$

$$= \lim_{x\to 0}\frac{\frac{1}{a^x+x}(a^x loga+1)}{1}$$

$$= \frac{(1.loga+1)}{1} \qquad ,[\because loge = 1]$$

$$= loga + loge$$

$$\therefore \lim_{x\to 0} logy = \log(ae)$$

$$\therefore \lim_{x\to 0} y = (ae)$$

$$\therefore \lim_{x\to 0}(a^x+x)^{\frac{1}{x}} = ae$$

Example 6: Evaluate $\lim_{x\to 0}\left(\frac{e^x+e^{2x}+e^{3x}}{3}\right)^{\frac{1}{x}}$, $(1^\infty\ form)$

Solution:

- $y=\left(\frac{e^x+e^{2x}+e^{3x}}{3}\right)^{\frac{1}{x}}$ **,** Now take log on both the sides,
- $logy = \frac{1}{x}log\left(\frac{e^x+e^{2x}+e^{3x}}{3}\right)$
- $\lim_{x\to 0} logy = \lim_{x\to 0}\frac{log\left(\frac{e^x+e^{2x}+e^{3x}}{3}\right)}{x}$, $\left(\frac{0}{0}form\right)$

$$= \lim_{x\to 0}\frac{log(e^x+e^{2x}+e^{3x})-log3}{x} \qquad ,\left(\frac{0}{0}form\right)$$

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$$= \lim_{x\to 0} \frac{\frac{1}{e^x+e^{2x}+e^{3x}}(e^x+2e^{2x}+3e^{3x})}{1}$$

$$= \frac{1}{(1+1+1)} . (1+2+3)$$

$$= \frac{6}{3}$$

$\therefore \lim_{x\to 0} logy = 2$

$\therefore \lim_{x\to 0} y = e^2$

$$\therefore \lim_{x\to 0} \left(\frac{e^x+e^{2x}+e^{3x}}{3}\right)^{\frac{1}{x}} = e^2$$

Example 7: Evaluate $\lim_{x\to 0} \left(\frac{1^x+2^x+3^x}{3}\right)^{\frac{1}{x}}$

Solution: $y = \left(\frac{1^x+2^x+3^x}{3}\right)^{\frac{1}{x}}$**,** Now take log on both the sides,

- $logy = \frac{1}{x} log\left(\frac{1^x+2^x+3^x}{3}\right)$
- $\lim_{x\to 0} logy = \lim_{x\to 0} \frac{log\left(\frac{1^x+2^x+3^x}{3}\right)}{x}$, $\left(\frac{0}{0} form\right)$

$$= \lim_{x\to 0} \frac{log(1^x+2^x+3^x)-log3}{x} \quad , \left(\frac{0}{0} form\right)$$

$$= \lim_{x\to 0} \frac{\frac{1}{1^x+2^x+3^x}(1^x log1+2^x log2+3^x log3)}{1}$$

$$= \frac{1}{(1+1+1)} . (\log 1.2.3)$$

$$= \frac{log6}{3}$$

$\therefore \lim_{x\to 0} logy = log6^{\frac{1}{3}}$

$$\therefore \lim_{x\to 0} \left(\frac{1^x+2^x+3^x}{3}\right)^{\frac{1}{x}} = 6^{\frac{1}{3}}$$

Example 7: Evaluate $\lim_{x\to 0} (cosx)^{cotx}$, $(1^\infty\ form)$

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Solution:

- $y = (cosx)^{cotx}$, Now take log on both the sides,
- $logy = cotx.logcosx$
- $\lim_{x\to 0} logy = \lim_{x\to 0} cotx.logcosx$, $(0 \times \infty form)$

$$= \lim_{x\to 0} \frac{logcosx}{tanx} \quad , \left(\frac{0}{0} form\right)$$

$$= \lim_{x\to 0} \frac{\frac{-sinx}{cosx}}{sec^2x}$$

$$= \lim_{x\to 0} -sinx.cosx$$

$$= -(0)(1)$$

$\therefore \lim_{x\to 0} logy = 0$

$\therefore \lim_{x\to 0} y = e^0 = 1$

$\therefore \lim_{x\to 0} (cosx)^{cotx} = e^0 = 1$

Example 8: Evaluate $\lim_{x\to 0} \left(\frac{tanx}{x}\right)^{\frac{1}{x}}$, $(1^\infty\ form)$

Solution:

- $y = \left(\frac{tanx}{x}\right)^{\frac{1}{x}}$, Now take log on both the sides,
- $logy = \frac{1}{x}.log\left(\frac{tanx}{x}\right)$
- $\lim_{x\to 0} logy = \lim_{x\to 0} \frac{log\left(\frac{tanx}{x}\right)}{x}$, $\left(\frac{0}{0} form\right)$

$$= \lim_{x\to 0} \frac{\frac{x}{tanx}\left\{\frac{x.sec^2x - tanx}{x^2}\right\}}{1} \quad , \left(\frac{0}{0} form\right)$$

$$= \lim_{x\to 0} \frac{x.sec^2x - tanx}{x^2} \quad , \left(\frac{0}{0}\right)$$

$$= \lim_{x\to 0} \frac{sec^2x + 2xsec^2x.tanx - sec^2x}{2x}$$

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$$= \lim_{x\to 0} \frac{2x sec^2 x.tanx}{2x}$$

$$= \lim_{x\to 0} sec^2 x.tanx$$

$$= sec^2(0).tan(0)$$

$$\therefore \lim_{x\to 0} logy = 0$$

$$\therefore \lim_{x\to 0} y = e^0 = 1$$

$$\therefore \lim_{x\to 0} \left(\frac{tanx}{x}\right)^{\frac{1}{x}} = e^0 = 1$$

Example 9: Evaluate $\lim_{x\to 0} \left(\frac{tanx}{x}\right)^{\frac{1}{x^2}}$, $(1^\infty\ form)$

Solution:

- $y = \left(\frac{tanx}{x}\right)^{\frac{1}{x^2}}$, Now take log on both the sides,
- $logy = \frac{1}{x^2}.log\left(\frac{tanx}{x}\right)$
- $\lim_{x\to 0} logy = \lim_{x\to 0} \frac{log\left(\frac{tanx}{x}\right)}{x^2}$, $\left(\frac{0}{0} form\right)$

$$= \lim_{x\to 0} \frac{\frac{x}{tanx}\left\{\frac{x.sec^2x - tanx}{x^2}\right\}}{2x} \quad , \left(\frac{0}{0} form\right)$$

$$= \frac{1}{2}\lim_{x\to 0} \frac{x.sec^2x - tanx}{x^3} \quad , \left(\frac{0}{0}\right)$$

$$= \frac{1}{2}\lim_{x\to 0} \frac{sec^2x + 2x sec^2x.tanx - sec^2x}{3x^2}$$

$$= \frac{1}{2}\lim_{x\to 0} \frac{2x sec^2x.tanx}{3x^2}$$

$$= \frac{1}{3}\lim_{x\to 0} sec^2x.\left(\frac{tanx}{x}\right)$$

$$= \frac{1}{3}(1)$$

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$$\therefore \lim_{x\to 0} logy = \frac{1}{3}$$

$$\therefore \lim_{x\to 0} y = e^{\frac{1}{3}}$$

$$\therefore \lim_{x\to 0} \left(\frac{tanx}{x}\right)^{\frac{1}{x^2}} = e^{\frac{1}{3}}$$

Example 10: Evaluate $\lim_{x\to a}\left(2-\frac{x}{a}\right)^{tan\left(\frac{\pi x}{2a}\right)}$, $(1^{\infty}\ form)$

Solution:

- $y = \left(2-\frac{x}{a}\right)^{tan\left(\frac{\pi x}{2a}\right)}$, Now take log on both the sides,
- $logy = tan\left(\frac{\pi x}{2a}\right) log\left(2-\frac{x}{a}\right)$
- $\lim_{x\to a} logy = \lim_{x\to a} \frac{log\left(2-\frac{x}{a}\right)}{cot\left(\frac{\pi x}{2a}\right)}$, $\left(\frac{0}{0}\ form\right)$

Take $\frac{x}{a} = t$

When $x \to a \Longrightarrow t \to 1$

$$= \lim_{t\to 1} \frac{log(2-t)}{cot\left(\frac{\pi}{2}t\right)} \quad , \left(\frac{0}{0}\ form\right)$$

$$= \lim_{t\to 1} \frac{\frac{-1}{(2-t)}}{-cosec^2\left(\frac{\pi}{2}t\right).\left(\frac{\pi}{2}\right)}$$

$$= \lim_{t\to 1} \frac{2}{\pi} \frac{sin^2\left(\frac{\pi}{2}t\right)}{(2-t)}$$

$$= \frac{2}{\pi}.\frac{1}{1}$$

$$\therefore \lim_{x\to 0} logy = \frac{2}{\pi}$$

$$\therefore \lim_{x\to a} y = e^{\frac{2}{\pi}}$$

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$$\therefore \lim_{x\to a}\left(2-\frac{x}{a}\right)^{tan\left(\frac{\pi x}{2a}\right)} = e^{\frac{2}{\pi}}$$

Example 11: Evaluate $\lim_{x\to 0}\left(\frac{1}{x}\right)^{tanx}$, $(\infty^0\ form)$

Solution:

- $y = \left(\frac{1}{x}\right)^{tanx}$ **,** Now take log on both the sides,
- $logy = tanx.log\left(\frac{1}{x}\right)$
- $\lim_{x\to 0} logy = \lim_{x\to 0} tanx.log\left(\frac{1}{x}\right)$, $(0\times\infty\ form)$

$$= \lim_{x\to 0}\frac{log1-logx}{cotx}$$

$$= \lim_{x\to 0}\frac{-logx}{cotx} \quad , \left(\frac{\infty}{\infty}\ form\right)$$

$$= \lim_{x\to 0}\frac{-\frac{1}{x}}{-cosec^2x}$$

$$= \lim_{x\to 0}\frac{\frac{1}{x}}{\frac{1}{sin^2x}}$$

$$= \lim_{x\to 0}\frac{sinx}{x}.sinx$$

$$= (1)(0)$$

$$\therefore \lim_{x\to 0} logy = 0$$

$$\therefore \lim_{x\to 0} y = e^0 = 1$$

$$\therefore \lim_{x\to 0}\left(\frac{1}{x}\right)^{tanx} = 1$$

Example 12: Evaluate $\lim_{x\to 0}\left(\frac{1}{x}\right)^{1-cosx}$, $(\infty^0\ form)$

Solution:

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- $y = \left(\frac{1}{x}\right)^{1-cosx}$, Now take log on both the sides,
- $logy = (1 - cosx).log\left(\frac{1}{x}\right)$
- $\lim_{x\to 0} logy = \lim_{x\to 0} (1 - cosx).log\left(\frac{1}{x}\right)$, $(0 \times \infty\ form)$

$$= \lim_{x\to 0} 2sin^2\frac{x}{2}.(log1 - logx) \quad , \left[\because 1 - cosx = 2sin^2\frac{x}{2}\right]$$

$$= \lim_{x\to 0} 2sin^2\frac{x}{2}.(-logx) \quad , (0 \times \infty\ form)$$

$$= \lim_{x\to 0} \frac{-2logx}{cosec^2\frac{x}{2}} \quad , \left(\frac{\infty}{\infty}\ form\right)$$

$$= \lim_{x\to 0} \frac{-\frac{2}{x}}{2cosec\frac{x}{2}\left(-cosec\frac{x}{2}.cot\frac{x}{2}\right).\frac{1}{2}}$$

$$= \lim_{x\to 0} \frac{2sin^3\frac{x}{2}}{x.cos\frac{x}{2}}$$

$$= \lim_{x\to 0} \left(\frac{tan\frac{x}{2}}{\frac{x}{2}}\right).sin^2\frac{x}{2}$$

$$= (1)(0)$$

$\therefore \lim_{x\to 0} logy = 0$

$\therefore \lim_{x\to 0} y = e^0 = 1$

$\therefore \lim_{x\to 0} \left(\frac{1}{x}\right)^{1-cosx} = 1$

Example 13: Evaluate $\lim_{x\to 0}(cotx)^{\frac{1}{logx}}$, $(\infty^0\ form)$

Solution:

- $y = (cotx)^{\frac{1}{logx}}$, Now take log on both the sides,
- $logy = \frac{1}{logx}.log(cotx)$

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- $\lim_{x\to 0} logy = \lim_{x\to 0} \frac{1}{logx}.log(cotx)$, $\left(\frac{\infty}{\infty} form\right)$

$$= \lim_{x\to 0} \frac{\frac{1}{cotx}(-cosec^2x)}{\frac{1}{x}}$$

$$= \lim_{x\to 0} \frac{-x}{\frac{cosx}{sinx}.sin^2x}$$

$$= \lim_{x\to 0} \frac{-2x}{2sinx.cosx}$$

$$= \lim_{x\to 0} -\left(\frac{2x}{sin2x}\right)$$

$$= \lim_{x\to 0}(-1)$$

$\therefore \lim_{x\to 0} logy = -1$

$\therefore \lim_{x\to 0} y = e^{-1} = \frac{1}{e}$

$\therefore \lim_{x\to 0}(cotx)^{\frac{1}{logx}} = \frac{1}{e}$

H.W.

Example 14: Evaluate $\lim_{x\to 0} x^{sinx}$ **ANS: 1**

Example 15: Evaluate $\lim_{x\to \frac{\pi}{2}} (cosx)^{cos^2x}$ **ANS: 1**

Example 16: Evaluate $\lim_{x\to 0} \left(\frac{1^x+2^x+3^x}{3}\right)^{\frac{1}{x}}$ **ANS: $6^{\frac{1}{3}}$**

Example 17: Evaluate $\lim_{x\to \infty} \left(\frac{a^{\frac{1}{x}}+b^{\frac{1}{x}}+c^{\frac{1}{x}}}{3}\right)^{3x}$ $\left[Hint: let \frac{1}{x} = y\right]$ **ANS: abc**

Example 18: Evaluate $\lim_{x\to \frac{\pi}{2}} (cosecx)^{tan^2x}$ **ANS: $e^{\frac{1}{2}}$**

Example 19: Evaluate $\lim_{x\to 0} (cotx)^{sinx}$ **ANS: 1**

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